

Network Connectivity Guide

Northgate Group — IT Service Desk Manual | Version 3.1 | Effective 1 January 2026 | Owner: IT Operations

1. Scope

This guide covers VPN setup and errors, wireless troubleshooting in the office and at home, name resolution problems, and how to request access to a blocked network port. It also gives the escalation route for network incidents.

2. VPN Setup

The VPN client is part of the standard build. To complete first-time setup, sign in with the company account, approve the multi-factor authentication prompt on the enrolled device, and accept the certificate installation when prompted. The connection profile is delivered automatically and should not be edited by hand.

The VPN uses split tunnelling: internal systems are routed through the tunnel and general internet traffic goes out over the local connection. Internal systems cannot be reached without the VPN active, including from the office guest network. Remote workers must have the VPN connected before opening any internal application, as required by the Remote Work Policy.

3. Common VPN Error Codes

- VPN-401 — Authentication failed. The password is wrong or has expired, or the multi-factor prompt was not approved in time. Check the account is not locked out before retrying.
- VPN-403 — Account not authorised for remote access. The remote access group membership is missing; raise a ticket.
- VPN-812 — Certificate expired or missing. The device certificate has passed its renewal date; reconnect to the office network or raise a ticket for reissue.
- VPN-1005 — Connection established but no internal systems reachable. Usually a name resolution problem; see the DNS section below.
- VPN-1104 — Tunnel drops repeatedly within minutes. Usually an unstable local connection or a home router blocking the protocol.

4. Wireless Troubleshooting

Two office networks are published. The corporate network requires the company account and is the only network that reaches internal systems directly. The guest network is isolated from internal systems and requires the VPN for any work use.

- Forget the network on the device and reconnect, which clears a stale profile and is the single most effective fix.
- Confirm the wireless adapter is enabled and not blocked by a function key or a hardware switch; fault code E-70 covers an adapter missing from the operating system.
- Move closer to an access point and test again; a weak signal often presents as an authentication failure rather than as poor speed.
- Test with a wired connection through the docking station to establish whether the fault is wireless-specific. If the dock itself is not detected, see fault code E-42 in the Laptop Issue Guide.
- At home, restart the router and test another device on the same network before raising a ticket, so the service desk can rule out the home connection.

5. DNS and Name Resolution

Symptoms of a name resolution problem include internal addresses failing while public websites load normally, or an internal site loading over the office network but not over the VPN. The first step is to disconnect and reconnect the VPN, which reapplies the correct search domains.

If the problem persists, flush the local resolver cache and restart the device. Employees must not change DNS server settings manually on a company device; manually configured public resolvers break access to internal systems and are reset automatically by policy at the next refresh.

6. Port Access Requests

Outbound traffic is restricted to a defined set of ports. A request to open an additional port is raised as an IT ticket with the request type Software Issue, naming the application in the software name field.

The request must state the destination host, the port number and protocol, whether access is needed from the office network, the VPN or both, and the business justification. Requests are reviewed by IT Operations and, where the destination is external, by the security team. Target turnaround is five business days. Temporary openings expire after 90 days unless renewed.

7. Escalation for Network Incidents

A fault affecting a single employee is raised as a normal IT ticket. A fault affecting an entire floor, office or site should be reported to the IT service desk by telephone as well as by ticket, so that it can be handled as an incident rather than a queued request.

Suspected outages are confirmed on the service status page before a ticket is raised. During a declared outage the service desk posts updates on the status page every 30 minutes and individual tickets raised for the same outage are linked to the incident rather than worked separately.